



Big Data Pipeline for Fraud Risk Analytics

PostgreSQL · Sqoop · HDFS · Hive · Spark SQL · Spark ML · Superset

TEAM 20

Introduction

\$12.5B

Fraud losses in 2024

Consumers reported losing more than \$12.5 billion to fraud, a 25% increase compared with the previous year.

38%

Reports with money loss

The FTC reports that 38% of fraud reports in 2024 involved actual money loss.

→ End-to-end pipeline

Ingest, store, analyze, model,
and visualize

→ Distributed processing

HDFS, Hive, and Spark on YARN

→ Dashboard-ready
outputs

Superset artifacts from Stage IV

Project Framing

Dataset

IEEE-CIS Fraud Detection — a real-world transaction dataset from Kaggle

<https://www.kaggle.com/c/ieee-fraud-detection>

Target Proxy

isFraud is used as a risk proxy, since no native churn or default label is available

- ① Using fraud as a proxy task is an intentional engineering decision — acknowledged upfront to prevent evaluation ambiguity.



Dataset and Input Data

Four Source Files

train_transaction.
CSV

Primary labeled
transactions with
isFraud target

train_identity.csv

Identity attributes
joined to training
transactions

test_transaction.c
SV

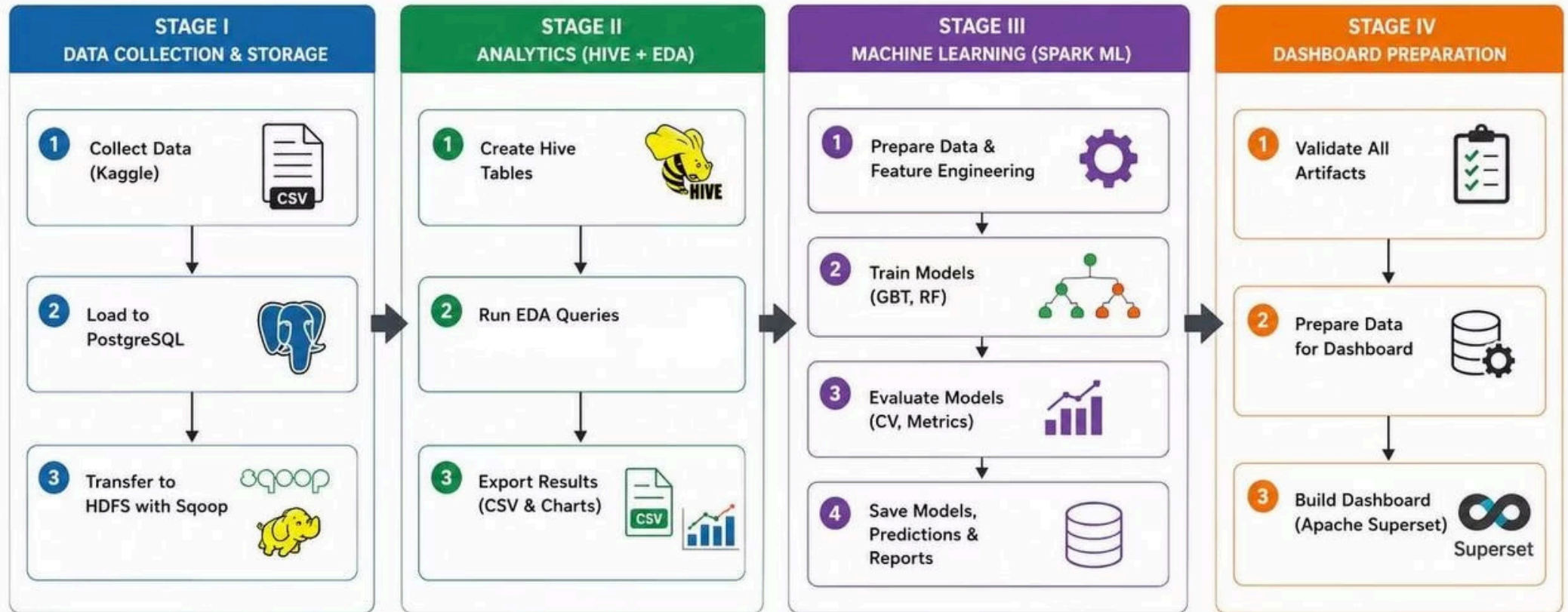
Unlabeled — used for
inference only, not
evaluation

test_identity.csv

Identity supplement for
test transactions

 Kaggle test files are **unlabeled**. All supervised evaluation is performed within the labeled training corpus.

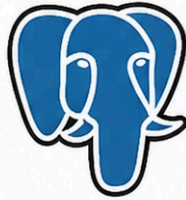
End-to-End Pipeline Overview



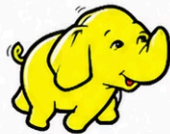
1 Collect Data
(Kaggle)



2 Load to
PostgreSQL



3 Transfer to
HDFS with Sqoop



Stage I — Data Collection & Storage

1

Validate CSVs

Schema checks and integrity validation on raw input files

2

Load to PostgreSQL

Structured relational storage as the ingestion landing zone

3

Transfer to HDFS

Sqoop export to HDFS as **Avro** + **Snappy** compressed format

1

Create Hive
Tables



2

Run EDA Queries



3

Export Results
(CSV & Charts)



Stage II — Hive Analytics & EDA

1

External Avro Tables

External Avro tables registered in Hive metastore from HDFS ingestion

2

ORC Analytics Table

Materialized ORC table optimized for fast aggregation and query speed

3

Hive Queries

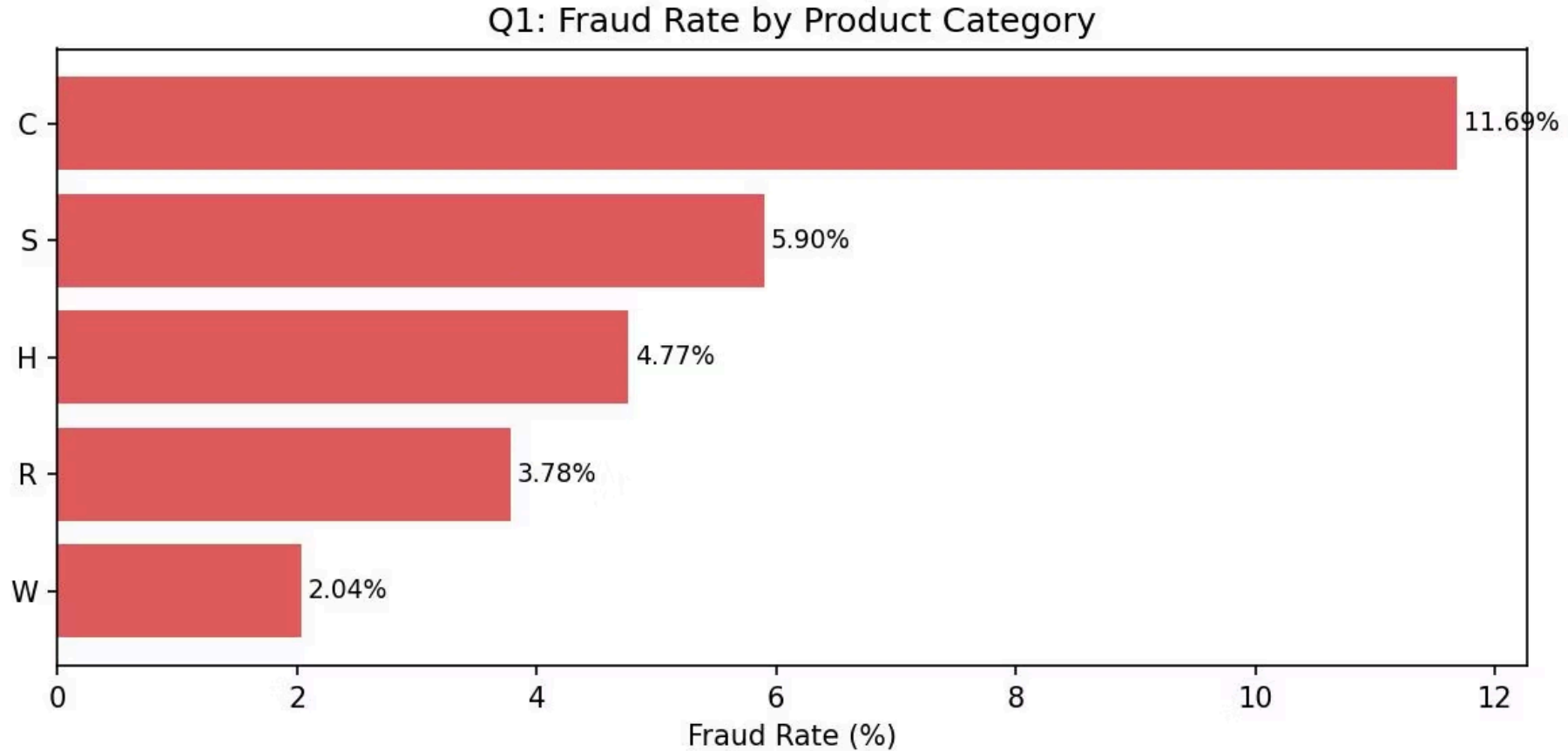
Five analytical queries run over the ORC table to extract fraud-risk signals

4

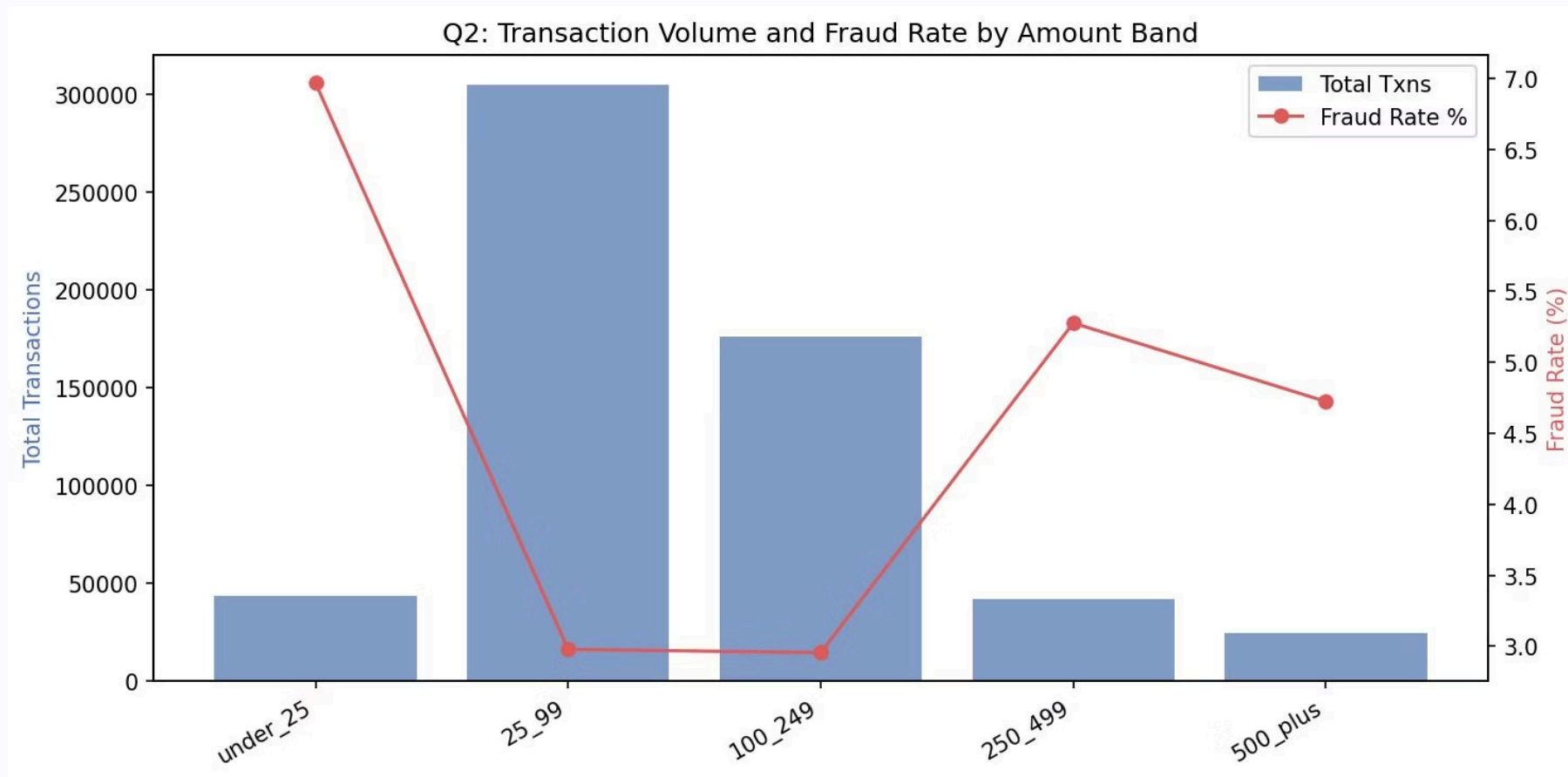
Export q1–q5 Charts & CSV

Results exported as chart images (q1–q5) and CSV summaries for EDA

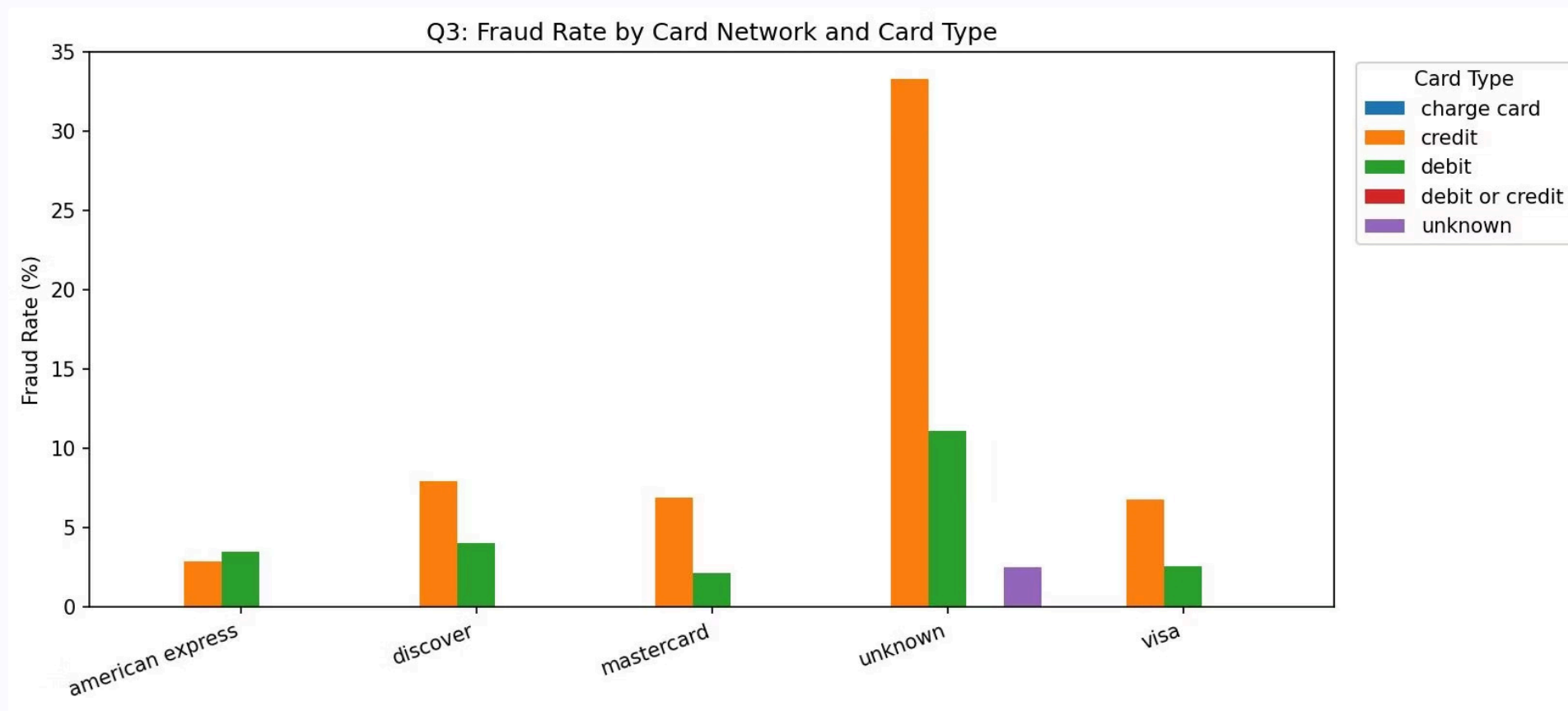
EDA Insight 1 — Product Risk



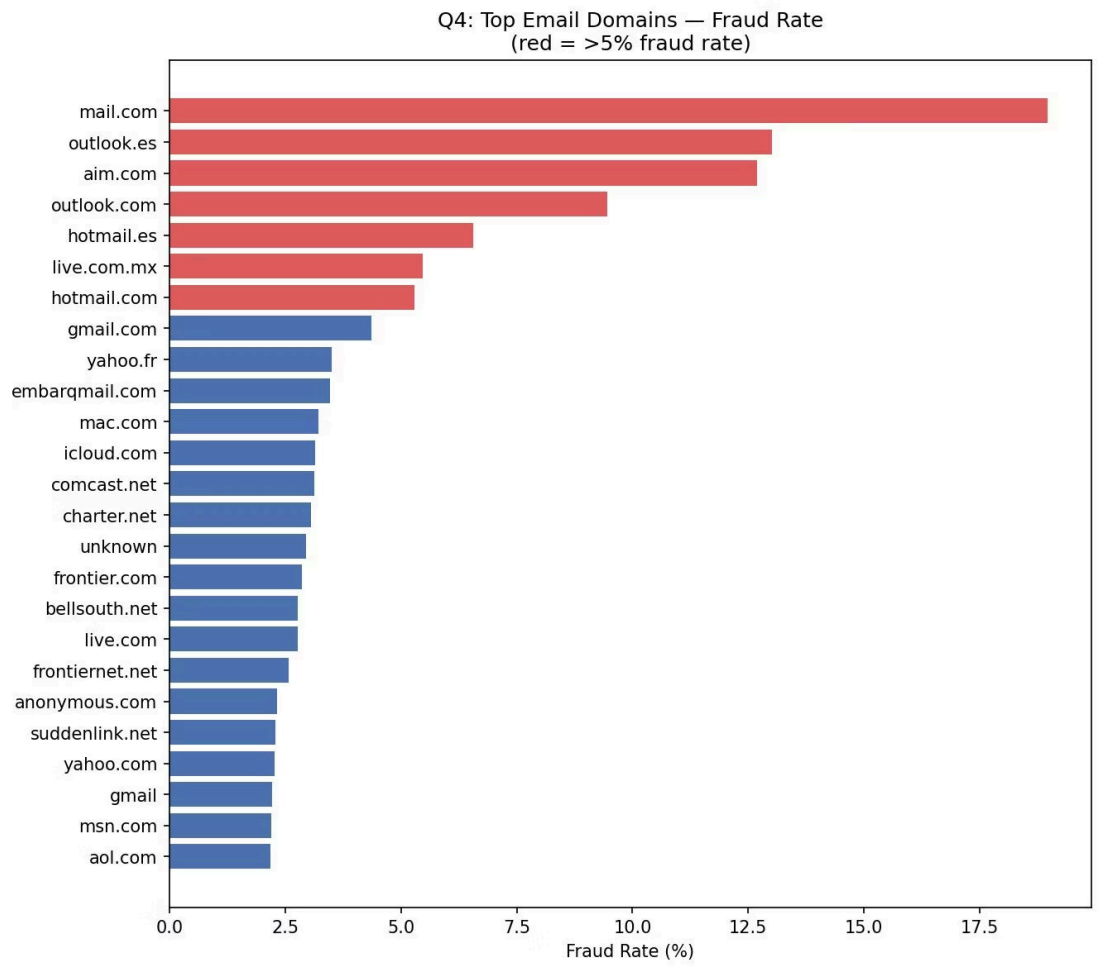
EDA Insight 1 — Amount Risk



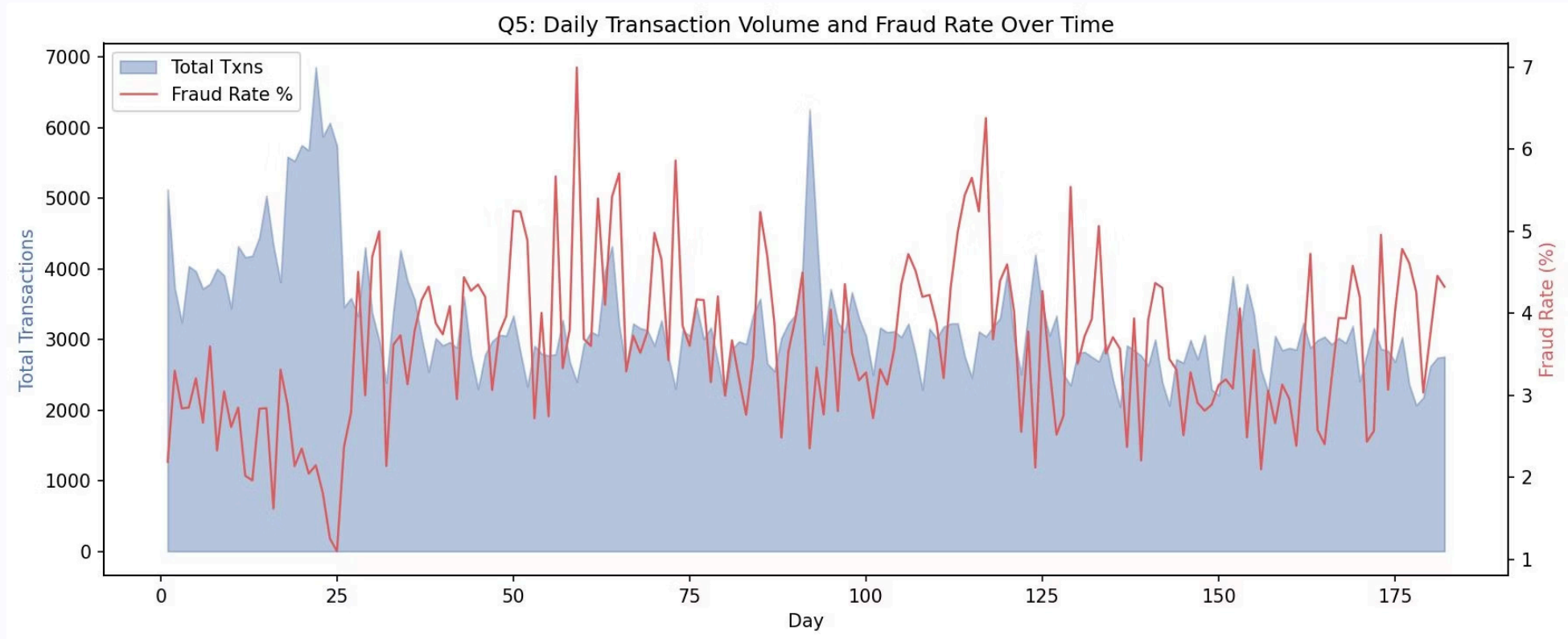
EDA Insight 2 — Card Type Risk



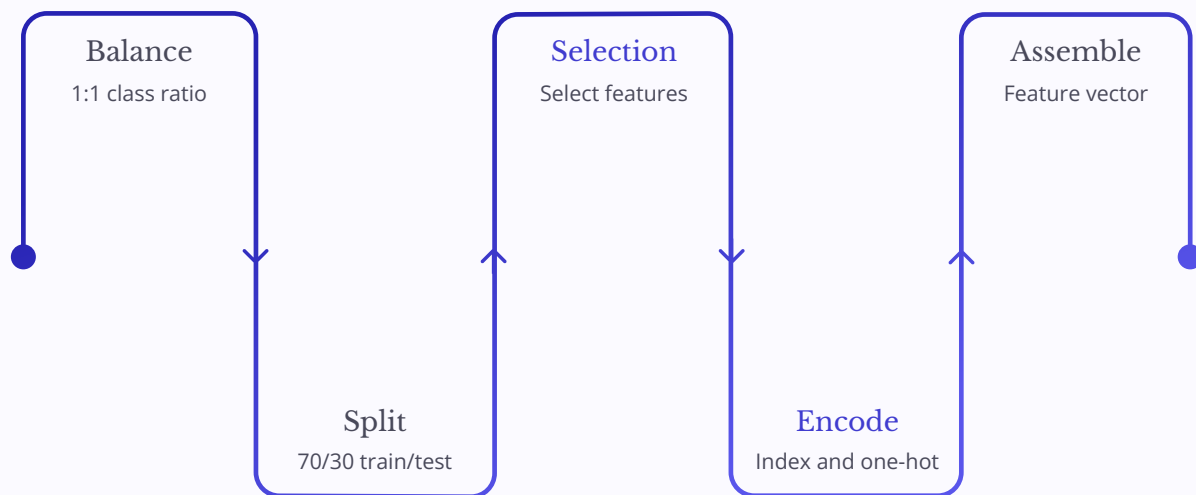
EDA Insight 2 — Email Domains



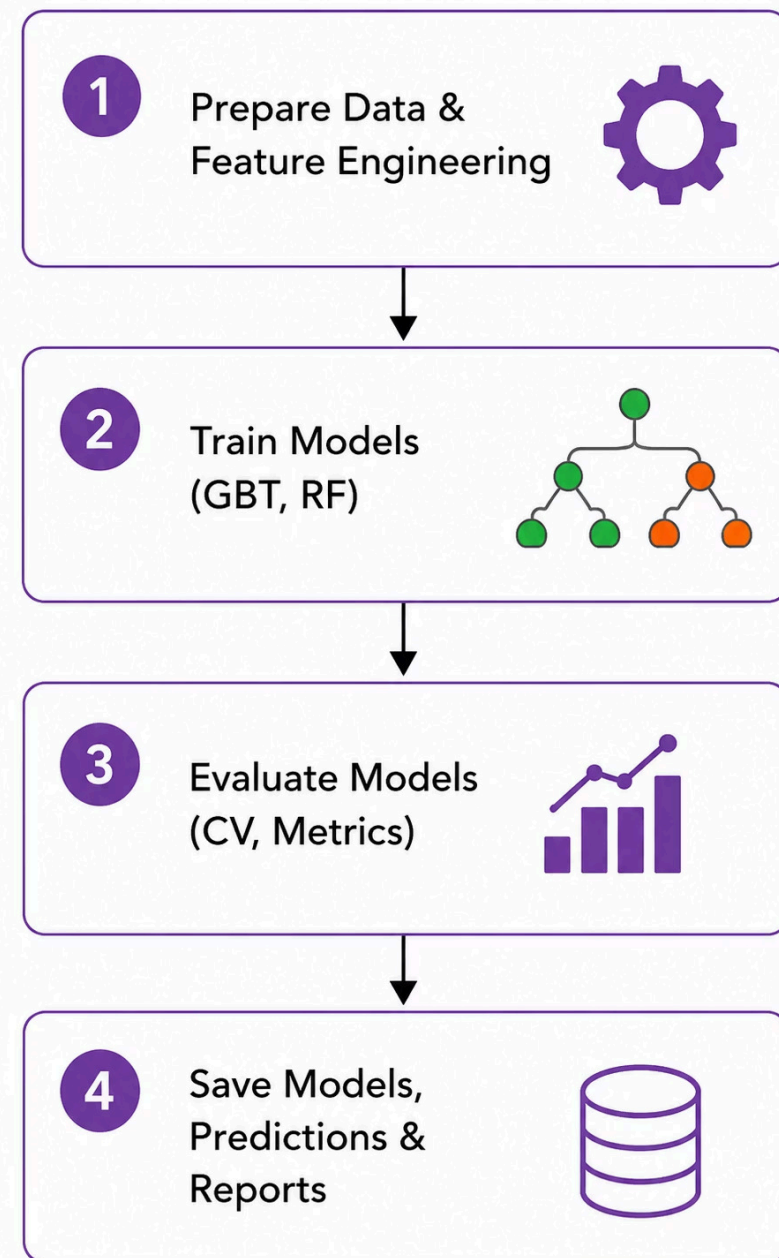
EDA Insight 2 — Daily Transactions Volume



Stage III — Distributed ML Pipeline



This is a full Spark ML pipeline executed in a distributed cluster environment. Every transformation is reproducible and scalable, with cross-validation guarding against overfitting.



Feature Engineering & Preprocessing

1

Remove Constant/Empty Features

Drop near-constant or empty features to reduce noise and dimensionality.

2

Handle String Features by Cardinality

Process categorical features based on their unique value count.

3

Importance-Based Pruning (Optional)

Conditionally filter features by their importance if enabled for optimization.

4

Clean Numeric Features

Convert to double data type and impute missing values using a median strategy.

5

Encode Categorical Features

Apply StringIndexer followed by OneHotEncoder for machine learning readiness.

6

VarianceThresholdSelector

Perform a final variance-based filtering to select informative features.

Model Comparison & Best Result

GBT — Best Model

Test AUC-PR: 0.932

Gradient Boosted Trees outperform Random Forest across every key evaluation metric. Selected as the final production model.

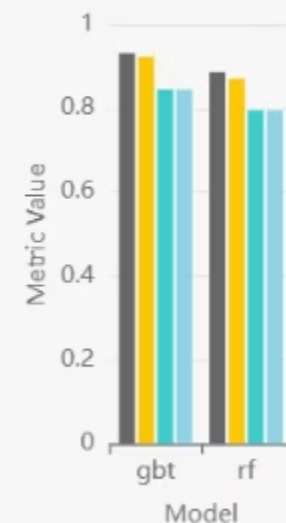
Random Forest — Baseline

A strong distributed baseline, but consistently outperformed by GBT on precision-recall and AUC metrics. Retained for comparison transparency.

- ✅ AUC-PR of **0.932** on a fraud dataset demonstrates strong model discrimination — well above a random baseline.

Model comparison

■ A ◀ 1/4 ▶ (All) (Inv)



DEMO

<http://hadoop-03.uni.innopolis.ru:8808/superset/dashboard/255/>

Limitations & Engineering Decisions

Known Constraints

→ Limited Business Interpretation

EDA reveals fraud-risk patterns, but without real customer profiles, credit history, and repayment behavior, we cannot draw strong conclusions about actual credit default.

Automatic Schema Inference Has Risk

The schema is generated from CSV headers and sample values. This is convenient for a wide dataset, but sampling may miss rare values. We reduced this risk by using a conservative fallback to `TEXT`

→ Cluster Resource Constraints

Feature filtering strategy was shaped by available distributed compute capacity



Conclusion

Built an End-to-End Big Data Pipeline

Ingestion → storage → analytics → ML → dashboard: all four stages connected into one system

Discovered Meaningful Fraud-Risk Patterns

EDA revealed actionable signals in product type, transaction amount, email domains, and time

Trained and Compared Distributed Spark ML Models

GBT selected as the best model with **Test AUC-PR: 0.932**, validated through 3-fold cross-validation